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TECHNICAL SPECIFICATION

Rail Telecommunications (RT);
FRMCS conformance specification;
Test Applicability and Implementation Conformance Statement (ICS) pro forma specification

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Foreword

This Technical Specification (TS) has been produced by ETSI Technical Committee Railway Telecommunications (RT).

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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1 Scope

The present document specifies the Test Case Applicability and the Implementation Conformance Statement (ICS) pro forma for FRMCS (Release 19). It provides recommended applicability for client and server functionalities (MCPTT/MCData features relevant to FRMCS), and the ICS items required to declare implementation capabilities.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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The following referenced documents are necessary for the application of the present document.

[1] ETSI TS 122 280 (V19.7.0): "LTE; 5G; Mission Critical Services Common Requirements (MCCoRe); Stage 1 (3GPP TS 22.280 version 19.7.0 Release 19)".

[2] ETSI TS 123 280 (V19.8.1): "LTE; Common functional architecture to support mission critical services; Stage 2 (3GPP TS 23.280 version 19.8.1 Release 19)".

[3] ISO/IEC 9646-7:1995: "Information technology — Open Systems Interconnection — Conformance testing methodology and framework; Part 7: Implementation Conformance Statements".

[4] ETSI TS 138 101-1 (V19.3.1): "5G; NR; User Equipment (UE) radio transmission and reception; Part 1: Range 1 Standalone (3GPP TS 38.101-1 version 19.3.1 Release 19)".

[5] ETSI TS 138 101-2 (V19.2.0): "5G; NR; User Equipment (UE) radio transmission and reception; Part 2: Range 2 Standalone (3GPP TS 38.101-2 version 19.2.0 Release 19)".

[6] ETSI TS 138 101-3 (V19.3.0): "5G; NR; User Equipment (UE) radio transmission and reception; Part 3: Range 1 and Range 2 Interworking operation with other radios (3GPP TS 38.101-3 version 19.3.0 Release 19)".

[7] ETSI TS 138 300 (V19.1.0): "5G; NR; NR and NG-RAN Overall description; Stage-2 (3GPP TS 38.300 version 19.1.0 Release 19)".

[8] ETSI TS 137 340 (V19.1.0): "Universal Mobile Telecommunications System (UMTS); LTE; 5G; Evolved Universal Terrestrial Radio Access (E-UTRA) and NR; Multi-connectivity; Overall Description; Stage-2 (3GPP TS 37.340 version 19.1.0 Release 19)".

[9] ETSI TS 124 501 (V19.5.0): "5G; Non-Access-Stratum (NAS) protocol for 5G System (5GS); Stage 3 (3GPP TS 24.501 version 19.5.0 Release 19)".

[10] ETSI TS 124 193 (V19.5.0): "5G; 5G System; Access Traffic Steering, Switching and Splitting (ATSSS); Stage 3; (3GPP TS 24.193 version 19.5.0 Release 19)".

[11] ETSI TS 138 331 (V19.0.0): "5G; NR; Radio Resource Control (RRC); Protocol specification (3GPP TS 38.331 version 19.0.0 Release 19)".

[12] ETSI TS 137 324 (V19.0.0): "LTE; 5G; Evolved Universal Terrestrial Radio Access (E-UTRA) and NR; Service Data Adaptation Protocol (SDAP) specification (3GPP TS 37.324 version 19.0.0 Release 19)".

[13] ETSI TS 138 323 (V19.0.0): "5G; NR; Packet Data Convergence Protocol (PDCP) specification (3GPP TS 38.323 version 19.0.0 Release 19)".

[14] ETSI TS 138 322 (V19.1.0): "5G; NR; Radio Link Control (RLC) protocol specification (3GPP TS 38.322 version 19.1.0 Release 19)".

[15] ETSI TS 138 321 (V19.0.0): "5G; NR; Medium Access Control (MAC) protocol specification (3GPP TS 38.321 version 19.0.0 Release 19)".

[16] ETSI TS 138 201 (V19.0.0): "5G; NR; Physical layer; General description (3GPP TS 38.201 version 19.0.0 Release 19)".

[17] ETSI TS 138 509 (V18.1.0): "LTE; 5G; 5GS; Special conformance testing functions for User Equipment (UE) (3GPP TS 38.509 version 18.1.0 Release 18)".

[18] ETSI TS 138 307 (V19.3.0): "5G; NR; Requirements on User Equipments (UEs) supporting a release-independent frequency band (3GPP TS 38.307 version 19.3.0 Release 19)".

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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The following referenced documents may be useful in implementing an ETSI deliverable or add to the reader's understanding but are not required for conformance to the present document.

[i.1] ETSI TS 104 069-1 (V0.0.21): "Rail Telecommunications (RT); Future Railway Mobile Communication System (FRMCS); Services over 5G; Part 1: Application conformance specification for client".

[i.2] ETSI TS 104 069-2 (V0.0.21): "Rail Telecommunications (RT); Future Railway Mobile Communication System (FRMCS); Services over 5G; Part 2: Application conformance specification for server".

[i.3] ETSI TR 121 905 (V18.0.0): "Digital cellular telecommunications system (Phase 2+) (GSM); Universal Mobile Telecommunications System (UMTS); LTE; 5G; Vocabulary for 3GPP Specifications (3GPP TR 21.905 version 18.0.0 Release 18)".

3 Definitions of terms, symbols and abbreviations

3.1 Terms

For the present document, the terms given in ETSI TS 122 280 [1], ETSI TS 123 280 [2] and the following apply:

ad hoc Group communication: combining of a multiplicity of MC service users into a group for the duration of a communication

NOTE: When the communication is released, the group no longer exists. If the communication is associated with an alert, then the group continues to exist until the alert is also cancelled.

ad hoc Group emergency alert: emergency alert addressed to multiple MC service users which are part of a group

NOTE: When the alert is cancelled, the group no longer exists. If the alert is associated with a communication, then the group continues to exist until the communication is also cancelled.

functional alias: user selectable alias that is tied to the assignment or task of the user

NOTE: A MCX User can activate one or multiple functional aliases at the same time. The activation of the functional alias(es) will take place after the user has signed in to the MCX Service using the MCX User ID.

group communication: one-to-many or many-to-many communication using an MCX Service

NOTE: The combination of different MC service users into a static group. MC service users have to affiliate for using the group service and need to de-affiliate in case they want to leave. In case of a communication, all affiliated MC service users will be involved.

MCX service emergency alert: notification from the MCX UE to the MCX Service that the MCX User has an emergency condition

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in ETSI TR 121 905 [i.3] and the following apply.

NOTE: An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in ETSI TR 121 905 [i.3].

FD	File Distribution
ICS	Implementation Conformance Statement
IXIT	Implementation eXtra Information for Testing
MC	Mission Critical
MCData	Mission Critical Data
MCPTT	Mission Critical Push To Talk
MCX	Mission Critical Services
SDS	Short Data Service

4 Recommended Test Case Applicability

The applicability of each individual test is identified in Table 4-1 (MCPTT Client) and Table 4-2 (MCPTT Server). This is just a recommendation based on the purpose for which the test case was written.

The applicability of every test is formally expressed using Boolean expression that are based on parameters (ICS) included in annex A of the present document.

Additional information related to the Test Case (TC), e.g. affecting its dynamic behaviour or its execution may be provided as well.

The columns in Table 4-1 and Table 4-2 have the following meaning:

Clause

The clause column indicates the clause number in ETSI TS 104 069-1 [i.1] or ETSI TS 104 069-2 [i.2] respectively which contains the test body.

Title

The title column describes the name of the test and contains the clause title of the clause in ETSI TS 104 069-1 [i.1] or ETSI TS 104 069-2 [i.2] respectively which contains the test body.

Release

The release column indicates the earliest release from which the test case is applicable. In some specific cases it may indicate the release(s) for which the TC is only applicable.

NOTE 1: Some exceptions to this interpretation may be indicated in Notes in column 'Number of TC Executions'.

Applicability - Condition

The following notations are used for the applicability column:

R	recommended - the test case is recommended
O	optional – the test case is optional
N/A	not applicable - in the given context, the test case is not recommended.
Ci	conditional - the test is recommended ("R") or not ("N/A") depending on the support of other items. "i" is an integer identifying a unique conditional status expression which is defined immediately following the table. For nested conditional expressions, the syntax "IF ... THEN (IF ... THEN ... ELSE...) ELSE ..." is used to avoid ambiguities.

NOTE 2: The conditions are defined in Table 4-1 (MCPTT Client) and Table 4-2 (MCPTT Server). To avoid ambiguity for the MCPTT Server testing conditions the notation of CCi is used.

Applicability - Comments

This column contains a verbal description of the condition.

Additional Information - Specific ICS

This column contains the mnemonics of ICS(s) affecting the dynamic behaviour of the TC.

NOTE 3: ICS items specified in other test specifications can be referred, to avoid redundant definitions.

Table 4-1: Client functionalities

Clause	TC Title	Release	Applicability		Additional Information		
			Condition	Comment	Specific ICS	Specific IXIT	Number of TC Executions
5.1	Functional aliasing client side	Rel-19					
5.2	Floor Control in general	Rel-19					
5.3	MCPTT Groups configured for multi-talker control	Rel-19					
5.4	Arbitration function	Rel-19					
6.1	Interception interface on MCX service system	Rel-19					
6.2	Ad hoc group communication functionality for MCX service user involving one MCX system	Rel-19					
6.3	Ad hoc group communication security	Rel-19					
6.4	Modifying the participants list of on-going ad hoc group communication	Rel-19					
6.5	Ad hoc group emergency alert support involving one MC system	Rel-19					
6.6	Ad hoc group call and ad hoc group emergency alert involving multiple MC systems	Rel-19					
6.7	MCData subservices SDS and FD support involving one MC system	Rel-19					
7.1	Mission critical location enhancements for on-network and off-network	Rel-19					
7.2	Preconfigured group enhancements	Rel-19					
7.3	Remotely initiated call request enhancements to support additional call setup parameters and implicit affiliation to the groups	Rel-19					
7.4	Geographical affiliation/de-affiliation	Rel-19					
7.5	Geographical sending of an emergency alert feature	Rel-19					
8.1	IP Connectivity	Rel-19					
8.2	MCData with different QoS requirements	Rel-19					
8.3	MCData streaming and datagram mode	Rel-19					
A1.1	Enhancements related to issues identified in any of the ETSI plug-tests for MC services	Rel-19					
A1.2	Support of MP-QUIC and MP-TCP	Rel-19					
A1.3	Support of IPv6	Rel-19					

Table 4-2: Server functionalities

Clause	TC Title	Release	Applicability		Additional Information		
			Condition	Comment	Specific ICS	Specific IXIT	Number of TC Executions
5.1	Functional aliasing server side	Rel-19					
5.2	Floor Control in general	Rel-19					
5.3	MCPTT Groups configured for multi-talker control	Rel-19					
6.1	Interception interface on MCX service system	Rel-19					
6.2	Administrative and configurational parameters required for Ad hoc group communications	Rel-19					
6.3	Ad hoc group communication functionality for MCX service user involving one MCX system	Rel-19					
6.4	Ad hoc group communication security	Rel-19					
6.5	Modifying the participants list of on-going ad hoc group communication	Rel-19					
6.6	Ad hoc group emergency alert support involving one MC system	Rel-19					
6.7	Configuration parameters required for Ad hoc group emergency alert support	Rel-19					
6.8	Interworking between MCX Service systems	Rel-19					
6.9	Ad hoc group call and ad hoc group emergency alert involving multiple MC systems	Rel-19					
6.10	MCData subservices SDS and FD support involving one MC system	Rel-19					
6.11	MCData subservices SDS and FD support involving multiple MC system	Rel-19					
6.12	Configuration parameters required for MCData subservices SDS and FD support	Rel-19					
7.1	Mission critical location enhancements for on-network and off-network	Rel-19					
7.2	Preconfigured group enhancements	Rel-19					
7.3	Remotely initiated call request enhancements to support additional call setup parameters and implicit affiliation to the groups	Rel-19					
7.4	Geographical affiliation/de-affiliation	Rel-19					
7.5	Geographical sending of an emergency alert feature	Rel-19					
8.1	IP Connectivity	Rel-19					

Clause	TC Title	Release	Applicability		Additional Information		
			Condition	Comment	Specific ICS	Specific IXIT	Number of TC Executions
8.2	MCData with different QoS requirements	Rel-19					
8.3	MCData streaming and datagram mode	Rel-19					
A.1	Support of MP-QUIC and MP-TCP	Rel-19					
A.2	Support of IPv6	Rel-19					

Annex A (normative): ICS pro forma for FRMCS Equipment

A.1 The right to copy

Notwithstanding the provisions of the copyright clause related to the text of the present document, ETSI grants that users of the present document may freely reproduce the ICS pro forma in this annex so that it can be used for its intended purposes and may further publish the completed ICS.

A.2 Guidance for completing the ICS pro forma

A.2.1 Purposes and structure

The purpose of this ICS pro forma is to provide a mechanism whereby a supplier of an implementation of the requirements defined in relevant specifications may provide information about the implementation in a standardized manner.

The ICS pro forma is subdivided into clauses for the following categories of information:

- instructions for completing the ICS pro forma.
- identification of the implementation.
- identification of the protocol.
- ICS pro forma tables (for example: UE implementation types, Teleservices, etc.).

A.2.2 Abbreviations and conventions

The ICS pro forma contained in this annex is comprised of information in tabular form in accordance with the guidelines presented in ISO/IEC 9646-7 [3].

Item column

The item column contains a number which identifies the item in the table.

Item description column

The item description column describes in free text each respective item (e.g. parameters, timers, etc.). It implicitly means "is <item description> supported by the implementation?".

Reference column

The reference column gives reference to the relevant 3GPP core specifications.

Release column

The release column indicates the earliest release from which the capability or option is introduced.

Mnemonic column

The Mnemonic column contains mnemonic identifiers for each item.

Comments column

This column is left blank for particular use by the reader of the present document.

References to items

For each possible item answer (answer in the support column) within the ICS pro forma there exists a unique reference, used, for example, in the conditional expressions. It is defined as the table identifier, followed by a solidus character "/", followed by the item number in the table. If there is more than one support column in a table, the columns shall be discriminated by letters (a, b, etc.), respectively.

A.2.3 Instructions for completing the ICS pro forma

The supplier of the implementation may complete the ICS pro forma in each of the spaces provided. More detailed instructions are given at the beginning of the different clauses of the ICS pro forma.

A.3 Identification of the User Equipment

A.3.1 Introduction

Identification of the User Equipment should be filled in so as to provide as much detail as possible regarding version numbers and configuration options.

The product supplier information and client information should both be filled in if they are different.

A person who can answer queries regarding information supplied in the ICS should be named as the contact person.

A.3.2 Date of the statement

.....

A.3.3 User Equipment Under Test (UEUT) identification

UEUT name:

.....

.....

Hardware configuration:

.....

.....

.....

Software configuration:

.....

.....

.....

A.3.4 Product supplier

Name:

.....

Address:

.....

.....

.....

Telephone number:

.....

Facsimile number:

.....

E-mail address:

.....

Additional information:

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.....

.....

A.3.5 Client

Name:

.....

Address:

.....

.....

.....

Telephone number:

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Facsimile number:

.....

E-mail address:

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Additional information:

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A.3.6 ICS contact person

Name:

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Telephone number:

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Facsimile number:

.....

E-mail address:

.....

Additional information:

.....

.....

A.4 Identification of the protocol

This ICS pro forma applies to the 3GPP standards listed in the normative references clause of the present document.

A.5 ICS pro forma tables

A.5.1 UE Implementation Types

Table A.5.1-1: UE Radio Technologies

Item	UE Radio Technologies	Reference	Release	Mnemonic	Comments
1	NR FDD	ETSI TS 138 101-1	Rel-19	pc_nrFDD	Support of band n100
2	NR TDD	ETSI TS 138 101-1 ETSI TS 138 101-2	Rel-19	pc_nrTDD	Support of band n101

Table A.5.1-2: UE general functionality

Item	UE Functionality	Reference	Release	Mnemonic	Comments
1	Multiple NR FDD bands	ETSI TS 138 101-1, clause 5.2	Rel-19	pc_nrFDD_MultiBand	
2	Multiple NR TDD bands	ETSI TS 138 101-1, clause 5.2 ETSI TS 138 101-2, clause 5.2	Rel-19	pc_nrTDD_MultiBand	

Table A.5.1-3: RAN-CN Interface Options

Item	UE support of RAN-CN Interface Options	Reference	Release	Mnemonic	Comments
1	NG-RAN NR	ETSI TS 138 300	Rel-19	pc_NG_RAN_NR	Option 2
2	EN-DC	ETSI TS 137 340	Rel-19	pc_EN_DC	Option 3
3	MR-DC with 5GC	ETSI TS 137 340	Rel-19	pc_NR_DC	

Table A.5.1-4: Inter-band NE-DC within FR1

Item	E-UTRA and NR DC Configuration	Reference	Release	Mnemonic	Comments
1	Spurious emission band UE co-existence	ETSI TS 138 101-3, clause 6.5B.3.3a.2	Rel-19	DC_n28_34	

Table A.5.1-5: 5GS UE Core Technologies

Item	5GS UE Core Technologies	Reference	Release	Mnemonic	Comments
1	UE Supports 5G Core Network	ETSI TS 124 501	Rel-19	pc_5GCN	
2	UE Supports only Stand-alone Non-Public Network	ETSI TS 124 501, clause 4.14 ETSI TS 138 300, clause 16.6.1	Rel-19	pc_SNPN_only	UEs operating only in SNPN access mode
3	UE Supports EPC over non-3GPP Access Network	ETSI TS 124 193, clause 4.8	Rel-19	pc_EPC_N3AN	

A.5.2 UE Service Capabilities

Table A.5.2-1: 5GS UE Service Capabilities

Item	5GS UE Core Technologies	Reference	Release	Mnemonic	Comments
1	Release of the access stratum minimum indication support for UE	ETSI TS 138 306	Rel-19	accessStratum Release	
2	Packet Data Convergence Protocol support	ETSI TS 138 306	Rel-19	Pdcp_Parameters	
3	Radio Link Control parameter support	ETSI TS 138 306	Rel-19	Rlc_Parameters	
4	Medium Access Control parameter support	ETSI TS 138 306	Rel-19	Mac_Parameters	
5	Physical parameter support	ETSI TS 138 306	Rel-19	Phy_Parameters	
6	Radio Frequency parameter support	ETSI TS 138 306	Rel-19	Rf_Parameters	
7	Measurement and mobility parameter support	ETSI TS 138 306	Rel-19	measAndMobParameters	
8	Supported feature sets	ETSI TS 138 306	Rel-19	featureSets	
9	Supported feature set combinations support	ETSI TS 138 306	Rel-19	featureSetCombinations (29)	
10	Noncritical extensions	ETSI TS 138 306	Rel-19	nonCriticalExtension	

A.5.3 Baseline Implementation Capabilities

A.5.3.1 Supported Protocols

Table A.5.3.1-1: Supported protocols

Item	Supported protocols	Reference	Release	Mnemonic	Comments
1	5GS Mobility Management	ETSI TS 124 501	Rel-19	5GmmSubLayerState	
2	5GS Session Management	ETSI TS 124 501	Rel-19	5GsmSubLayerState	
3	Radio Resource Control	ETSI TS 138 331	Rel-19	RRC layer	
4	Service Data Adaptation Protocol	ETSI TS 137 324	Rel-19	Sdap	
5	Packet Data Convergence Protocol	ETSI TS 138 323	Rel-19	Pdcp	
6	Radio Link Control	ETSI TS 138 322	Rel-19	RLC	
7	Medium Access Control	ETSI TS 138 321	Rel-19	MAC	
8	Physical Layer	ETSI TS 138 201	Rel-19	Phy	

A.5.3.2 Special Conformance Testing Functions

Table A.5.3.2-1: Special Conformance Testing Functions

Item	Special Conformance Testing Functions	Reference	Release	Mnemonic	Comments
1	UE test loop	ETSI TS 138 509	Rel-15		
2	UE Power Limit Function (UPLF)	ETSI TS 138 509, clause 5.11	Rel-16		

A.5.3.3 RF Baseline Implementation Capabilities

NOTE 1: The values indicated in column "Release" for bands are to be understood as the specifications release version in which a frequency band was introduced. It is not a mandate that a UE, corresponding to that specific release, support a particular band. For further guidance to release independent bands see ETSI TS 138 307 [18].

NOTE 2: See Annex B for status of completed NR bands and power classes in this version of 3GPP UE conformance test specifications.

Table A.5.3.3-1: NR FDD FR1 RF Baseline Implementation Capabilities

Item	NR FDD FR1 RF Baseline Implementation Capabilities	Reference	Release	Mnemonic	Comments
1	NR Frequency band: 874,4 MHz to 880 MHz (UL), 919,4 MHz to 925 MHz (DL)	ETSI TS 138 101-1, clause 5.2	Rel-19	pc_nrBand100_Supp	NR FDD FR1 Band n100
2	NR Frequency band: 1 900 MHz to 1 910 MHz	ETSI TS 138 101-1, clause 5.2	Rel-19	pc_nrBand101_Supp	NR TDD FR1 Band n101
3	NR Frequency band: 703 MHz to 748 MHz (UL), 758 MHz to 803 MHz (DL)	ETSI TS 138 101-1, clause 5.2	Rel-19	pc_nrBand28_Supp	NR FDD FR1 Band n28

Annex B (informative): Change history

Date	Version	Information about changes
23.01.2026	0.0.2	Early Draft
23.02.2026	0.0.18	Stable Draft
22.05.2026	0.0.19	Revised Stable Draft
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